

Capture the Flag 7: Stop Putting Everything on One Screen

🚩 Flag 0 — Create Your Feature Branch

Objective: Create a new feature branch before adding multiple screens.

Create:

`feature/multi-screen-navigation`

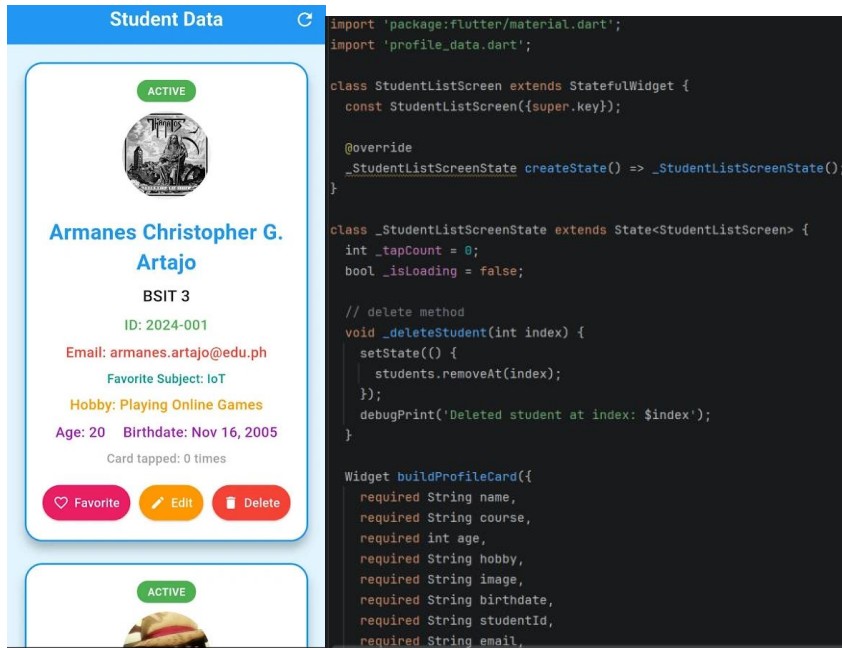
Screenshot:

```
C:\Users\Manes Christopher\Documents\mannes\IT314AB_Artajo>git branch
feature/conditional-ui
* feature/multi-screen-navigation
feature/student-interactions
```

🚩 Flag 1 — Create the Student List Screen

Objective: Move your Student Directory into its own screen.

Screenshot:



```
import 'package:flutter/material.dart';
import 'profile_data.dart';
import 'student_list_screen.dart';

void main() {
  students.sort((a, b) => a.name.compareTo(b.name));
  runApp(const MyApp());
}

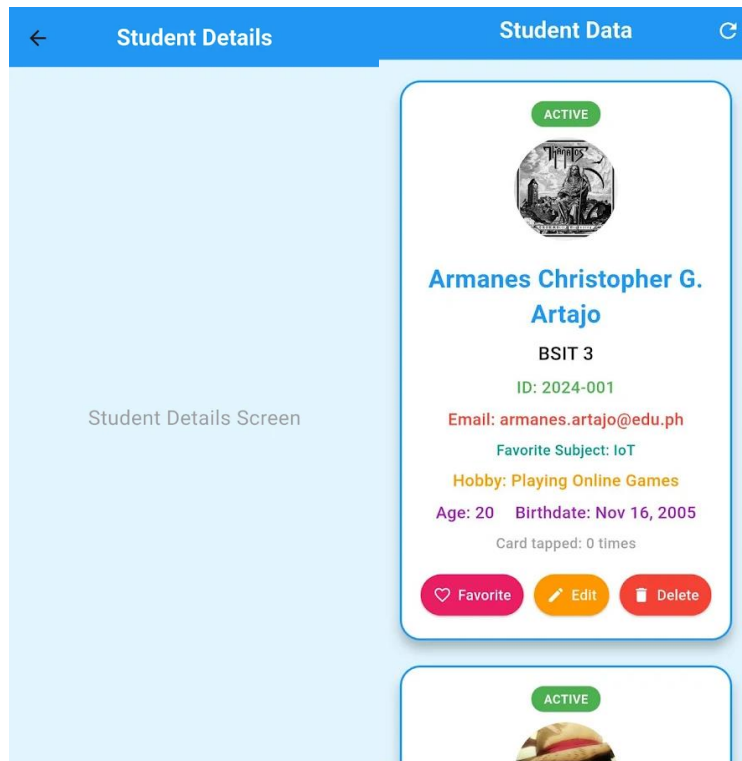
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: appTitle,
      home: const StudentListScreen(),
    );
  }
}
```

🚩 Flag 2 — Create the Student Details Screen

Objective: Create a separate screen for viewing a student's information.

Screenshot:



```
return GestureDetector(  
  onTap: () {  
    Navigator.push(  
      context,  
      MaterialPageRoute(  
        builder: (context) => const StudentDetailsScreen(),  
      ), MaterialPageRoute  
    );  
  },  
);
```

🚩 Flag 3 — Pass Student Data

Objective: Pass the selected student's data to the Student Details Screen.

Screenshot:



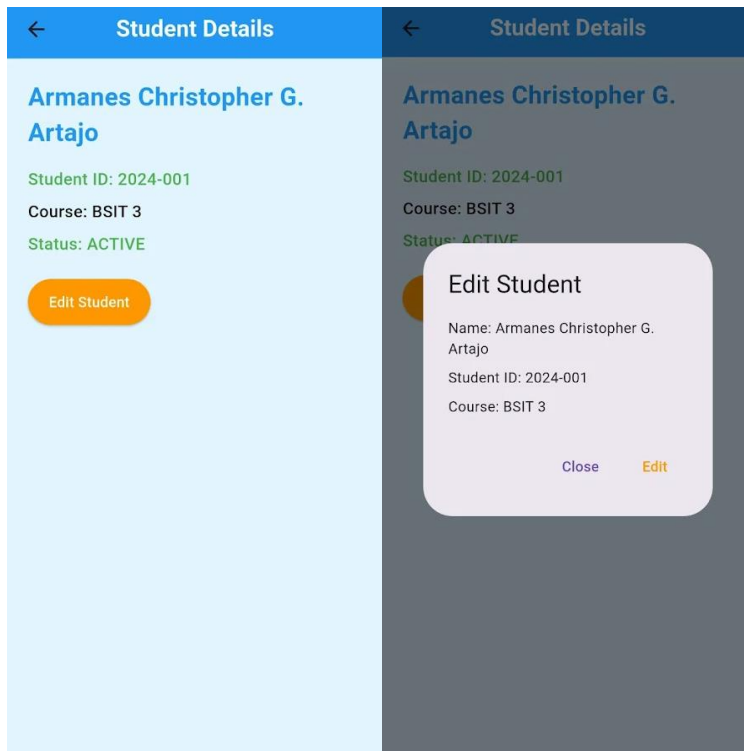
```
onTap: () {  
  Navigator.push(  
    context,  
    MaterialPageRoute(  
      builder: (context) => StudentDetailsScreen(student: student),  
    ),  
    MaterialPageRoute  
  );  
}
```

🚩 Flag 4 — Edit with a Modal

Objective: Allow administrators to edit student information without creating another screen.

Screenshot:

```
void showEditDialog(BuildContext context) {  
  showDialog(  
    context: context,  
    builder: (context) {  
      return AlertDialog(  
        title: const Text('Edit Student'),  
        content: Column(  
          mainAxisAlignment: MainAxisAlignment.min,  
          crossAxisAlignment: CrossAxisAlignment.start,  
          children: [  
            Text('Name: ${student.name}'),  
            const SizedBox(height: 8),  
            Text('Student ID: ${student.studentId}'),  
            const SizedBox(height: 8),  
            Text('Course: ${student.course}'),  
          ],  
        ),  
      );  
    },  
  );  
}
```



Flag 5 — Create the Add Student Screen

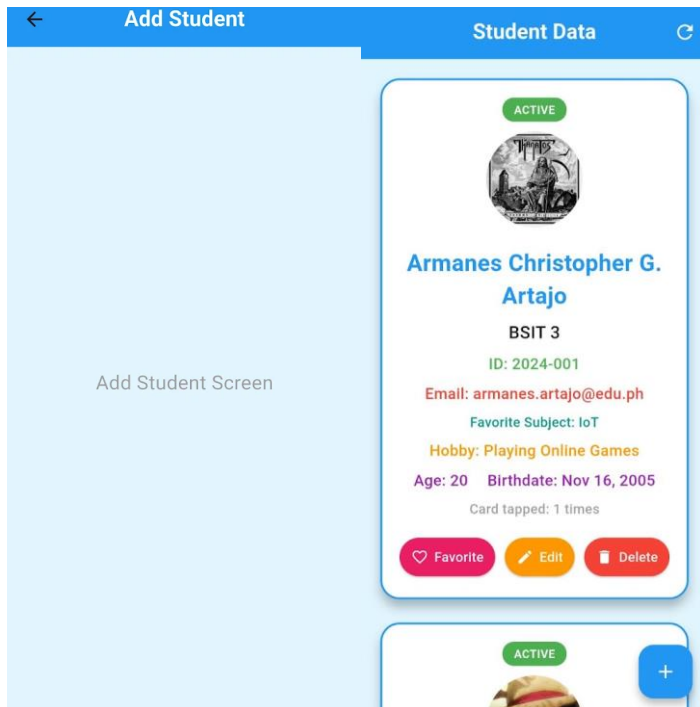
Screenshot:

```
import 'package:flutter/material.dart';

class AddStudentScreen extends StatelessWidget {
  const AddStudentScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: Colors.lightBlue[50],
      appBar: AppBar(
        backgroundColor: Colors.blue,
        centerTitle: true,
        title: const Text(
          'Add Student',
          style: TextStyle(
            color: Colors.white,
            fontWeight: FontWeight.bold,
          ),
        ),
      ),
      body: const Center(
        child: Text(
          'Add Student Screen',
          style: TextStyle(
            fontSize: 20,
            color: Colors.grey,
          ),
        ),
      ),
    );
  }
}

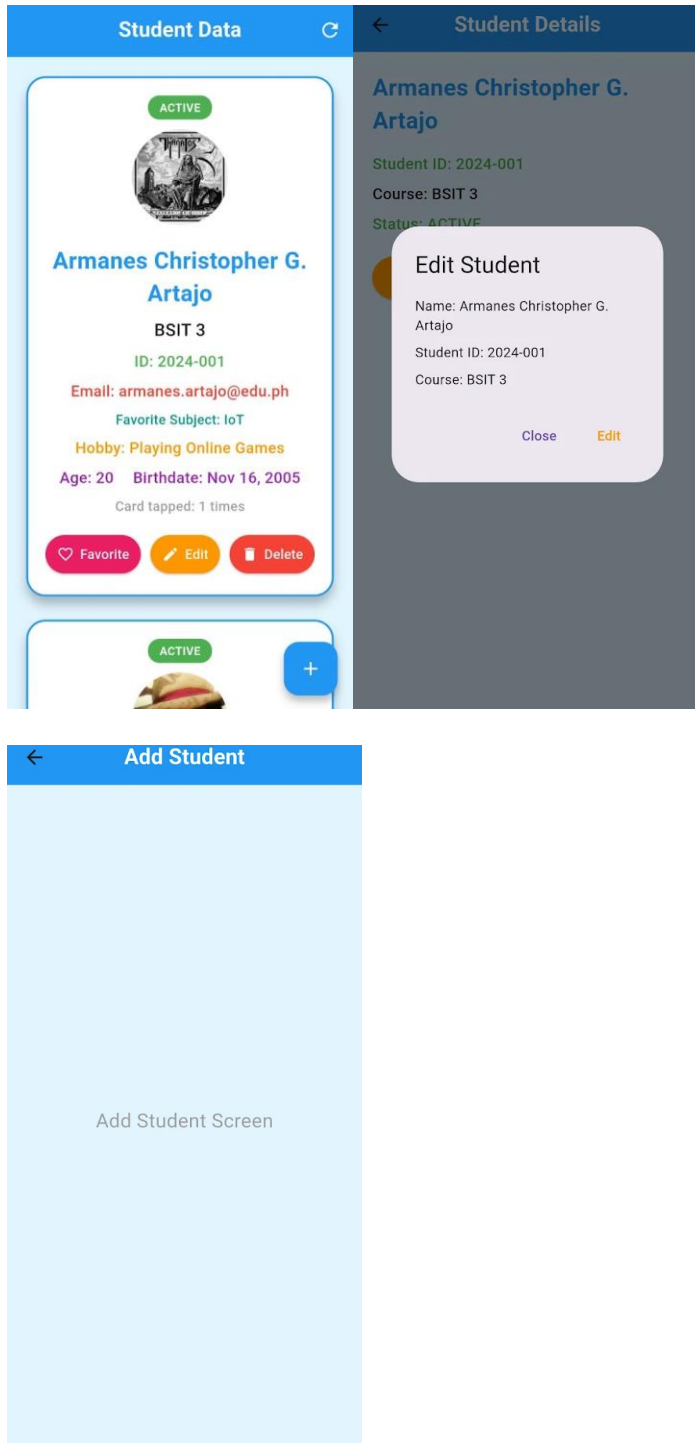
floatingActionButton: FloatingActionButton(
  onPressed: () {
    Navigator.push(
      context,
      MaterialPageRoute(
        builder: (context) => const AddStudentScreen(),
      ),
    );
  },
);
```



Flag 6 — Prepare the Add Student Screen

Objective: Prepare the Add Student Screen for future functionality.

Screenshot:



🚩 Flag 7 — Connect the Routes

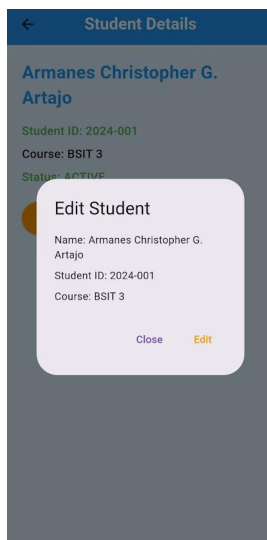
Objective: Organize the application's screens using Flutter routes.

Screenshot:

```
title: appTitle,  
initialRoute: '/student-list',  
routes: {  
  '/student-list': (context) => const StudentListScreen(),  
  '/student-details': (context) {  
    final student = ModalRoute.of(context)!.settings.arguments as Student;  
    return StudentDetailsScreen(student: student);  
  },  
  '/add-student': (context) => const AddStudentScreen(),  
}
```

```
Navigator.pushNamed(  
  context,  
  '/student-details',  
  arguments: student,  
);
```

```
floatingActionButton: FloatingActionButton(  
  onPressed: () {  
    Navigator.pushNamed(context, '/add-student');  
  },  
  backgroundColor: Colors.blue,  
  child: const Icon(Icons.add, color: Colors.white),  
),
```



🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Multi-Screen Student Directory.

Use a proper Git commit message.

Branch:

`feature/multi-screen-navigation`

Screenshot:

```
C:\Users\Manes Christopher\Documents\mannes\IT314AB_Artajo>git push origin feature/multi-screen-navigation
git: 'credential-manager-core' is not a git command. See 'git --help'.
Enumerating objects: 22, done.
Counting objects: 100% (22/22), done.
Delta compression using up to 4 threads
Compressing objects: 100% (16/16), done.
Writing objects: 100% (17/17), 4.14 MiB | 1.95 MiB/s, done.
Total 17 (delta 8), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (8/8), completed with 4 local objects.
remote:
remote: Create a pull request for 'feature/multi-screen-navigation' on GitHub by visiting:
remote:   https://github.com/BosconianManes/IT314AB_Artajo/pull/new/feature/multi-screen-navigation
remote:
To https://github.com/BosconianManes/IT314AB_Artajo.git
 * [new branch]      feature/multi-screen-navigation -> feature/multi-screen-navigation
```